

Course Name: DSM VLSI

Course Number and Section: **Year: Spring 2024**

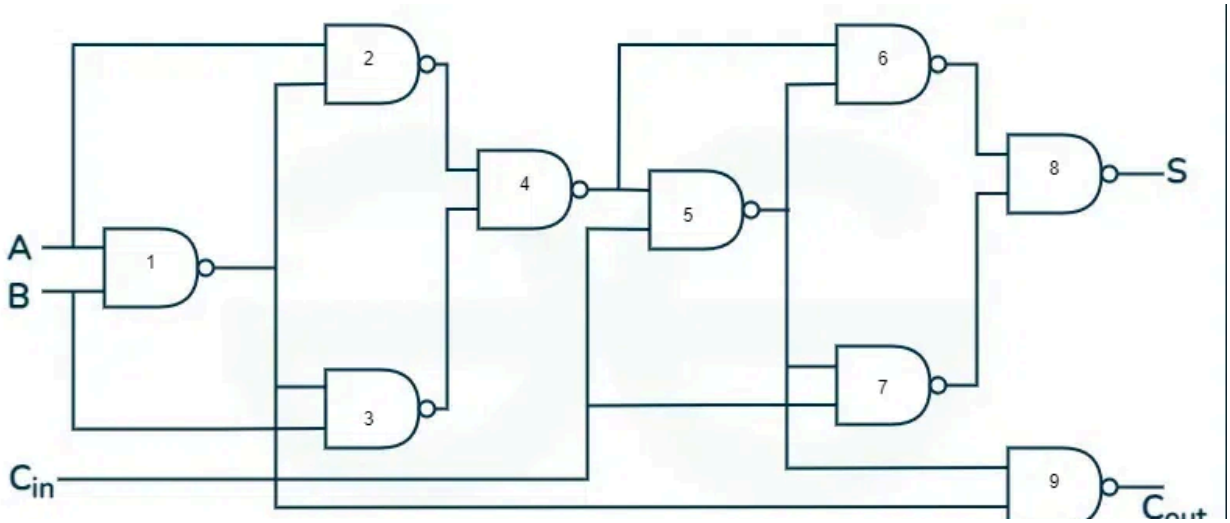
Student Name and RUID: Gautham Sathyanarayanan 223008966

Date Submitted: 4/15/2026

Initially, a 1 bit full adder was created. I decided to use a 2 input nand-nand implementation to simplify the logic gate w/l ratios while also allowing for the layout to be simpler in its connections due to repeating geometry. The logic for a 1 bit FA is as follows:

Sum = A ^ B ^ Cin ; Cout = AB + Cin(A^B)

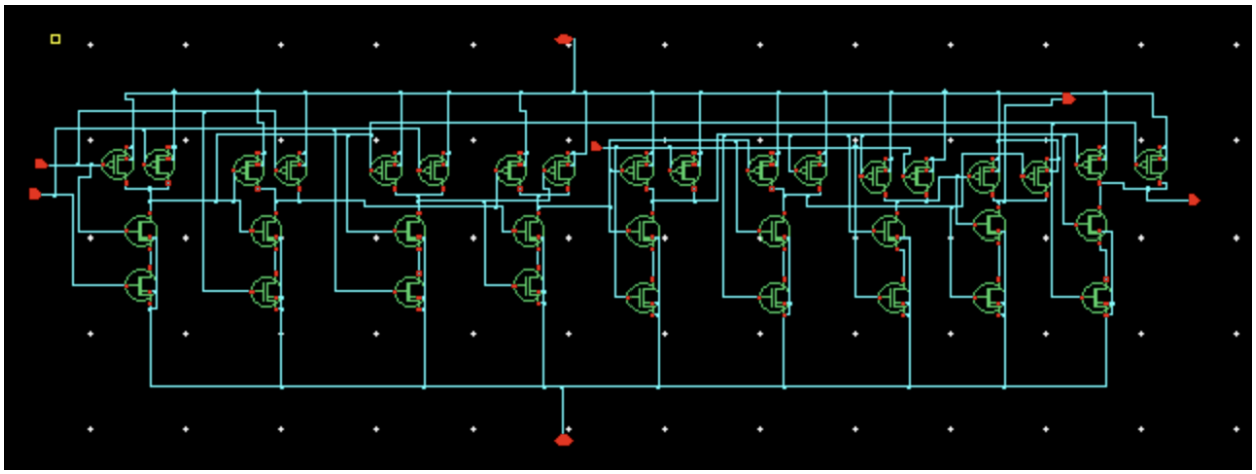
Since the both of them can have some gates reused, the total amount of nand gates is 9 in the following formation:

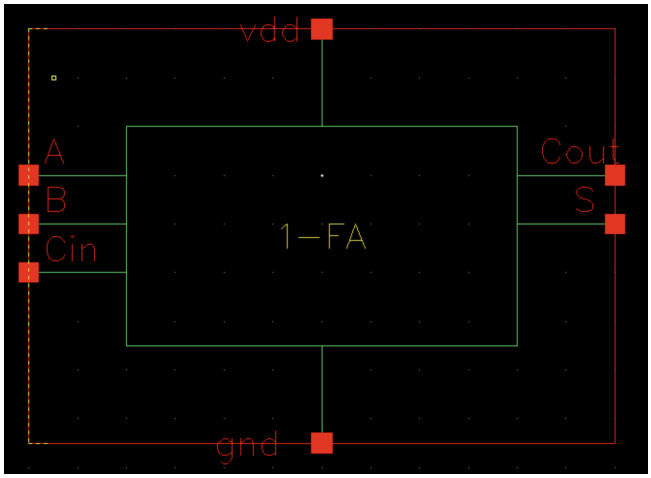


Logical Effort:

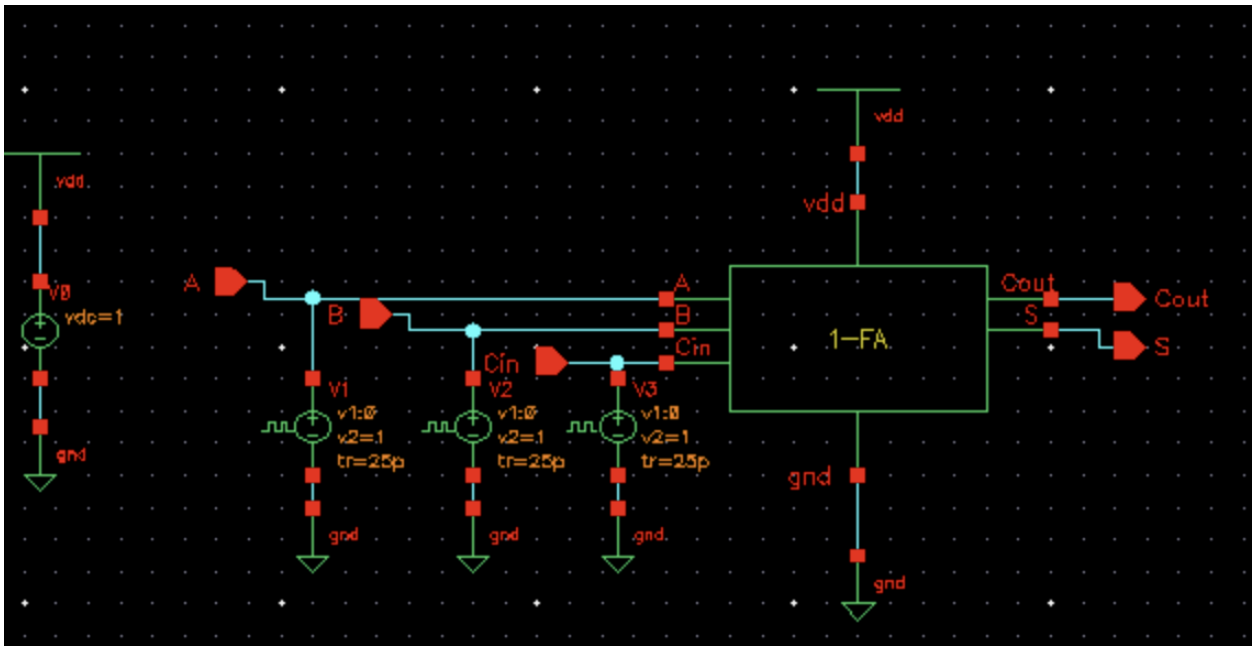
There are four main paths that the data can take, with A/B and Cin changing Sum and Cout respectively. The effort of A/B changing S is 6 *(nand g) vs Cin changing S is 3 (nand g). Consequently, A/B changing Cout is 2*(nand g) and Cin changing C out being 1(nand g). In normal nmos g of nand is 4/3, making the values evaluate to 24/3, 12/3, 8/3, and 4/3.

a normal nand gate was implemented for the following design:





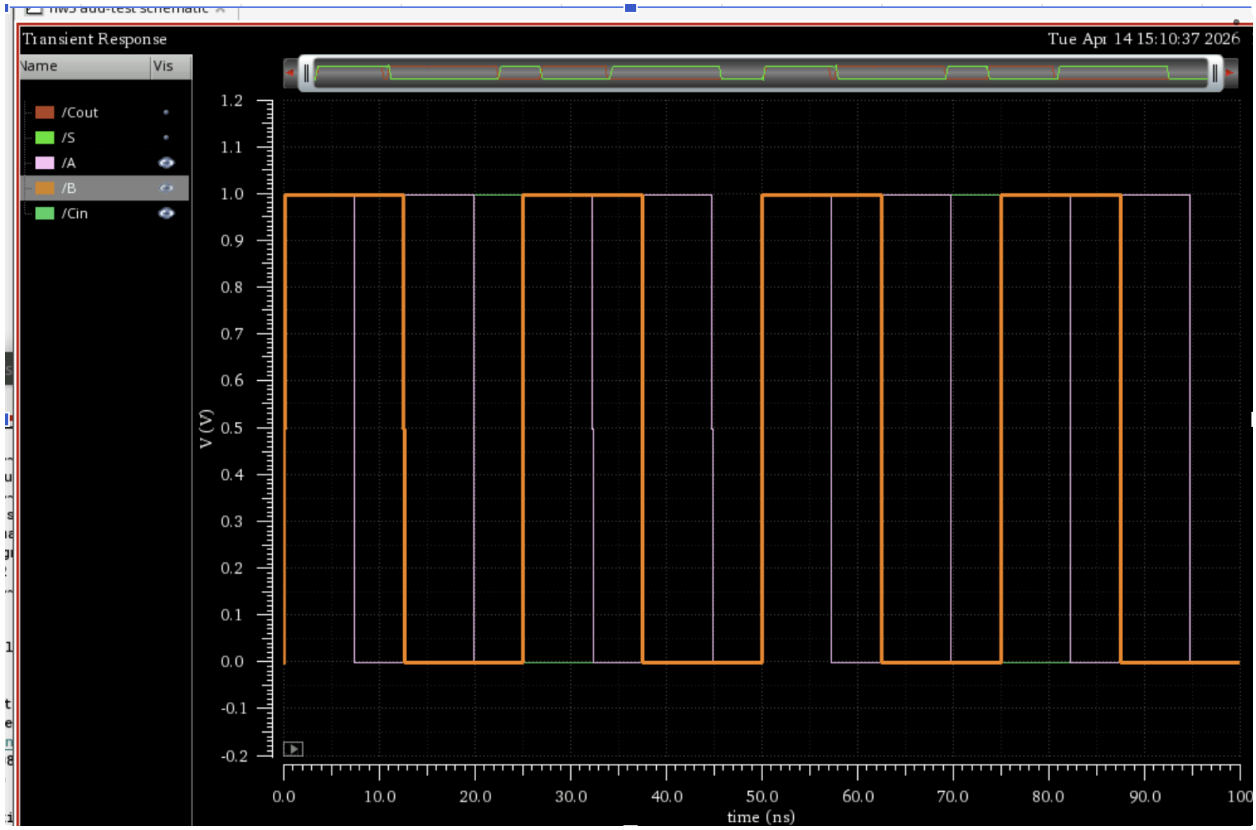
with the following test setup:



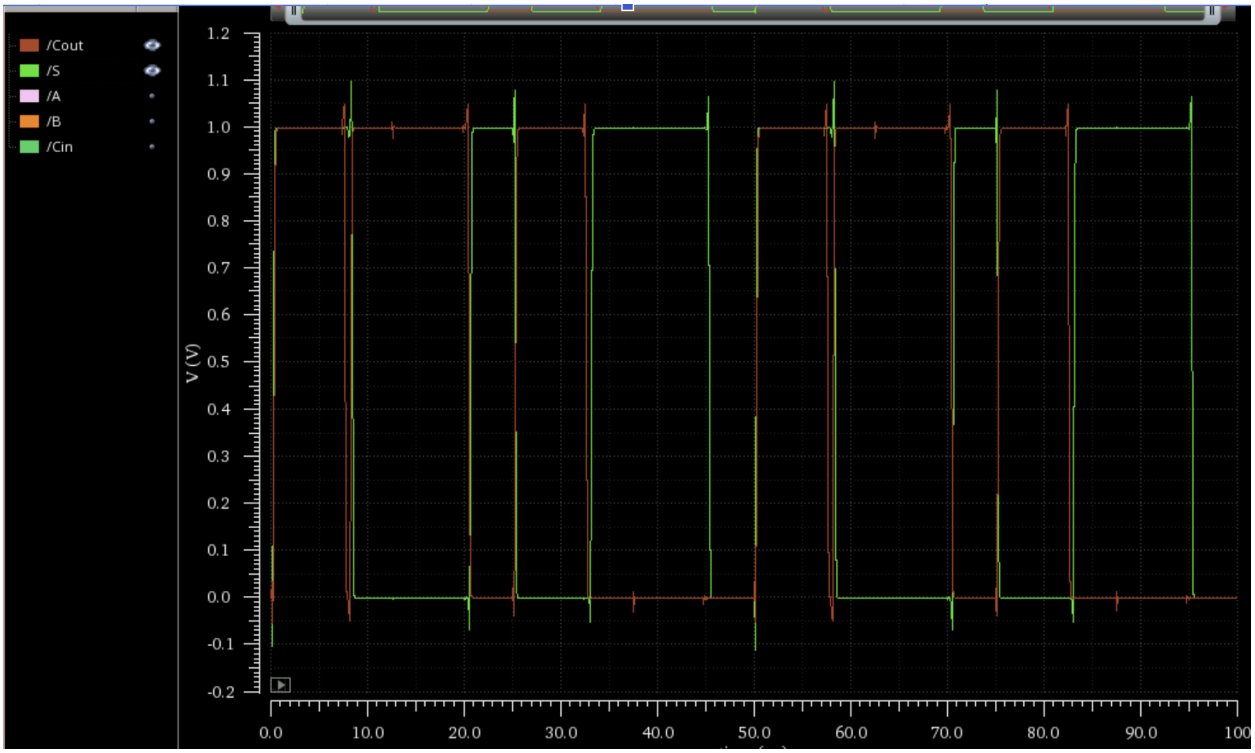
The rise and fall time of the input sources were 25 pico seconds. The on and off times were varied over multiple tests to get propagation delays.

First, every input combination was pulsed for the same amount of time to calculate average power and functionality.

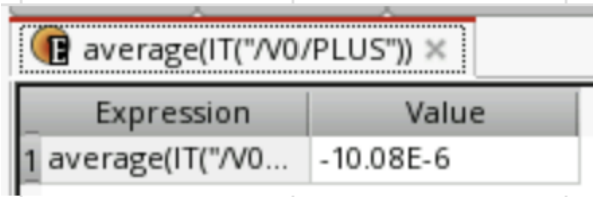
Inputs:


Transient Response
Tue Apr 14 15:10:37 2026
Name
Vis
/Cout
/S
/A
/B
/Cin
V (V)
time (ns)

Outputs:


/Cout
/S
/A
/B
/Cin
V (V)

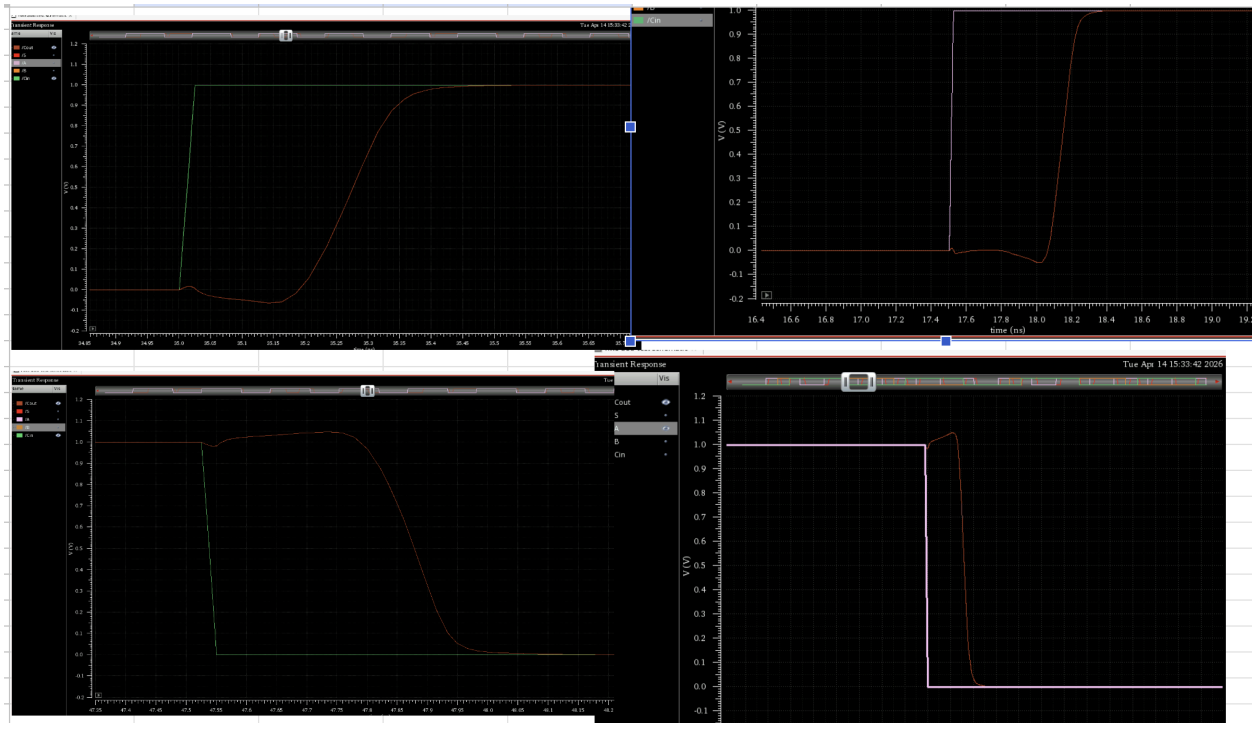
The average power was found by finding the average current coming out of the 1 volt power source, yielding 10.08 u W.

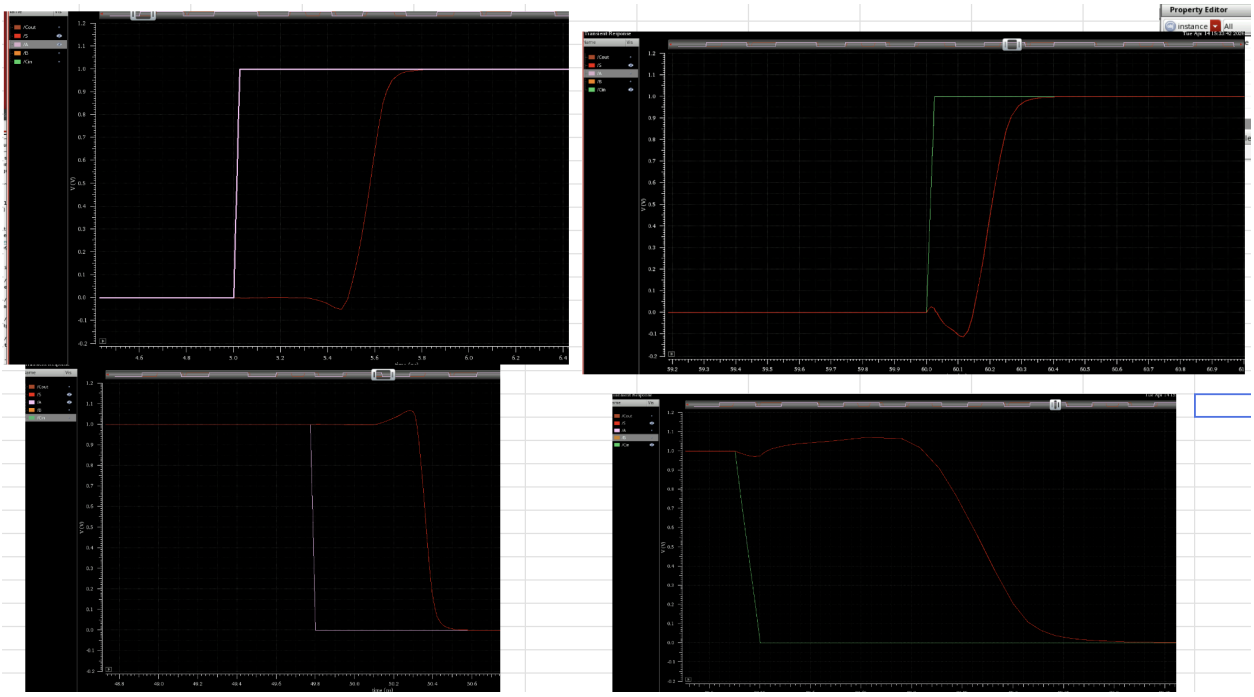


After trying different delays/periods, the following delays were derived: (ALL IN NANOSECONDS)

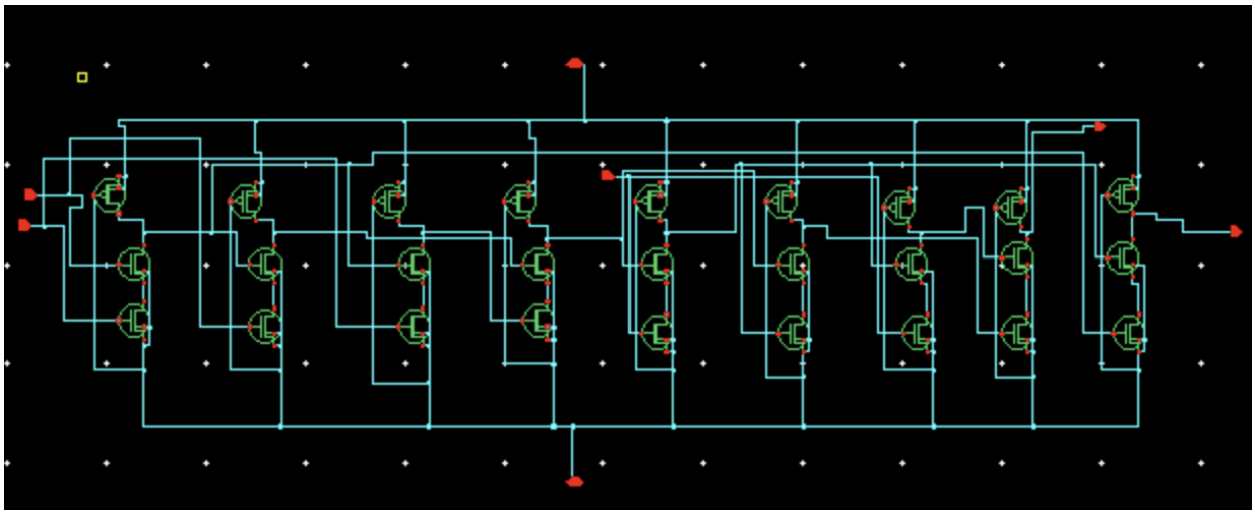
cout	tplh	tphl	tpd
A/B	0.278661	0.3296755	0.30416825
Cin	0.262838	0.340201	0.3015195

sum	tplh	tphl	tpd
A/B	0.56724	0.577919	0.5725795
Cin	0.193682	0.233469	0.2135755

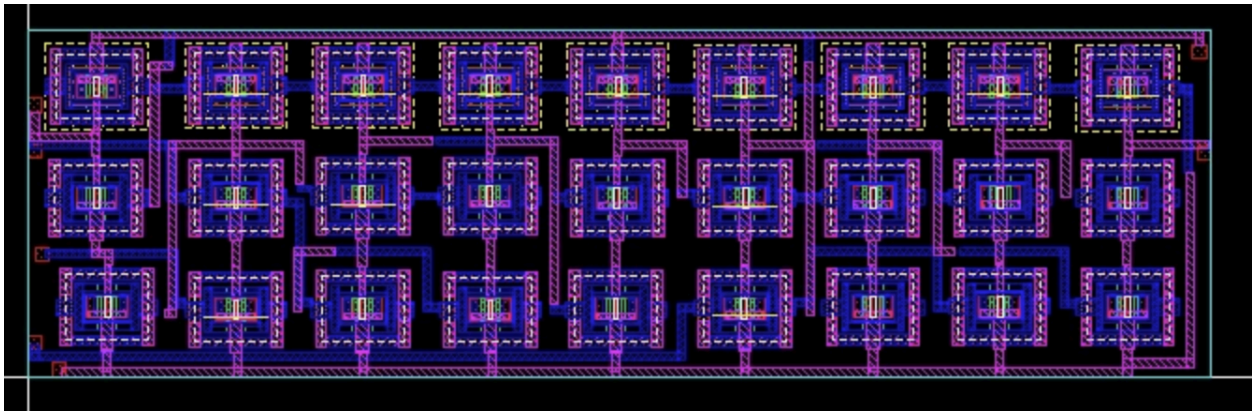




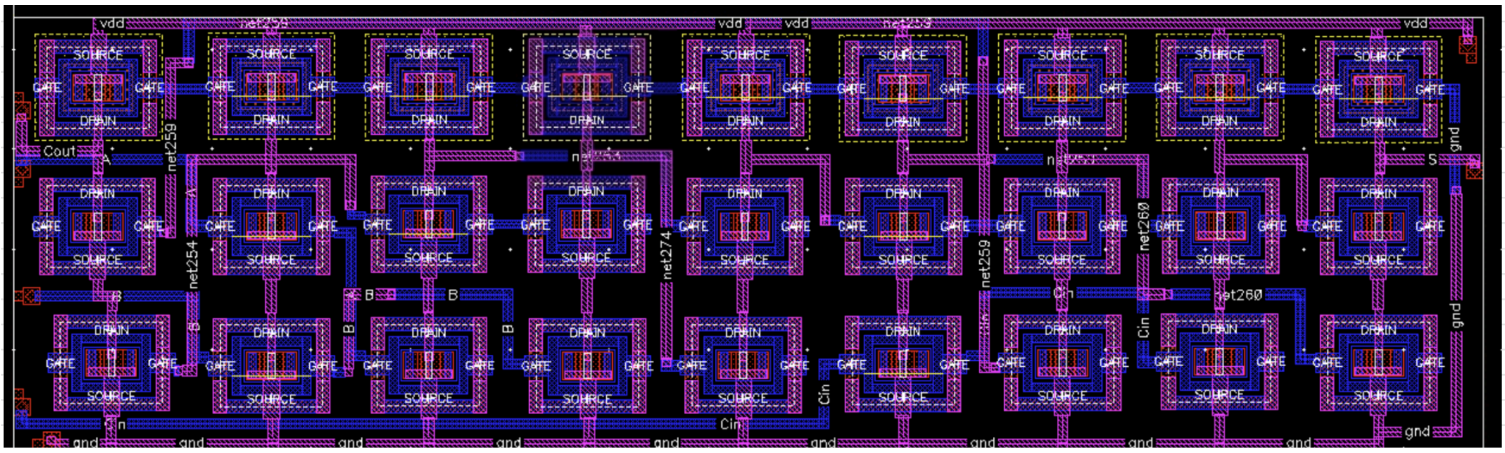
At this point, a layout had been started, but not using pseudo nmos was causing significant complexity that outweighed the theoretical benefits of using cmos, so a pseudo nmos design was changed to.



The main benefit derived from using pseudo nmos is the ability to stack nand gates to the sides of each other. When using a standard nmos, there is added width with the second nmos input per nand gate. That extra nmos causes the layout to not be symmetric, thus creating wasted space. In the following layout, the top mosfet represents the gate's grounded nmos, while the two pmos underneath represent the input of said gate.



The following image is closer up, but the top right metal interconnect is slightly offplaced. (This was fixed after)



The transistors are spaced such that 12 lambda, or 12*90nm = 1080nm. This is to satisfy the 4 lambda spacing practice used to avoid cross coupling/contamination when a wire is taking up the middle 4 lambda. The dimensions of the physical layout is 53.28 nm by 15.54 nm, taking an area of 827.9712 nm^2.